

RISK AND RETURN

RETURN FROM A STOCK

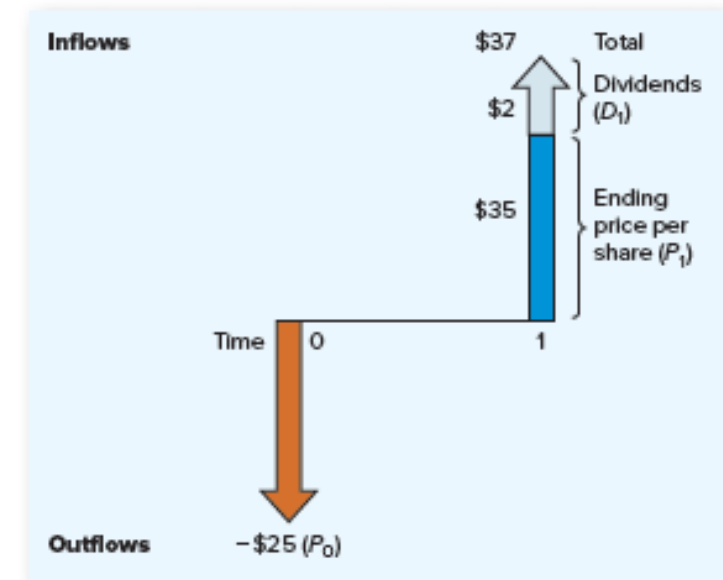
Your \$2 dividend per share works out to a dividend yield of:

$$\begin{aligned}\text{Dividend yield} &= D_{t+1}/P_t \\ &= \$2/\$25 = .08, \text{ or } 8\%\end{aligned}$$

The per-share capital gain is \$10, so the capital gains yield is:

$$\begin{aligned}\text{Capital gains yield} &= (P_{t+1} - P_t)/P_t \\ &= (\$35 - 25)/\$25 \\ &= \$10/\$25 \\ &= .40, \text{ or } 40\%\end{aligned}$$

The total percentage return is 48 percent.



$$\text{Total dollar return} = \text{Dividend income} + \text{Capital gain (or loss)}$$

ARITHMETIC VERSUS GEOMETRIC AVERAGES

- The geometric average return: “What was your average compound return per year over a particular period?”
- The geometric average tells you **what you actually earned per year on average, compounded annually**
- The arithmetic average return : “What was your return in an average year over a particular period?”
- The arithmetic average tells you **what you earned in a typical year.**

$$\text{Geometric average return} = [(1 + R_1) \times (1 + R_2) \times \dots \times (1 + R_T)]^{1/T} - 1$$

**Geometric versus
Arithmetic Average
Returns: 1926–2016**

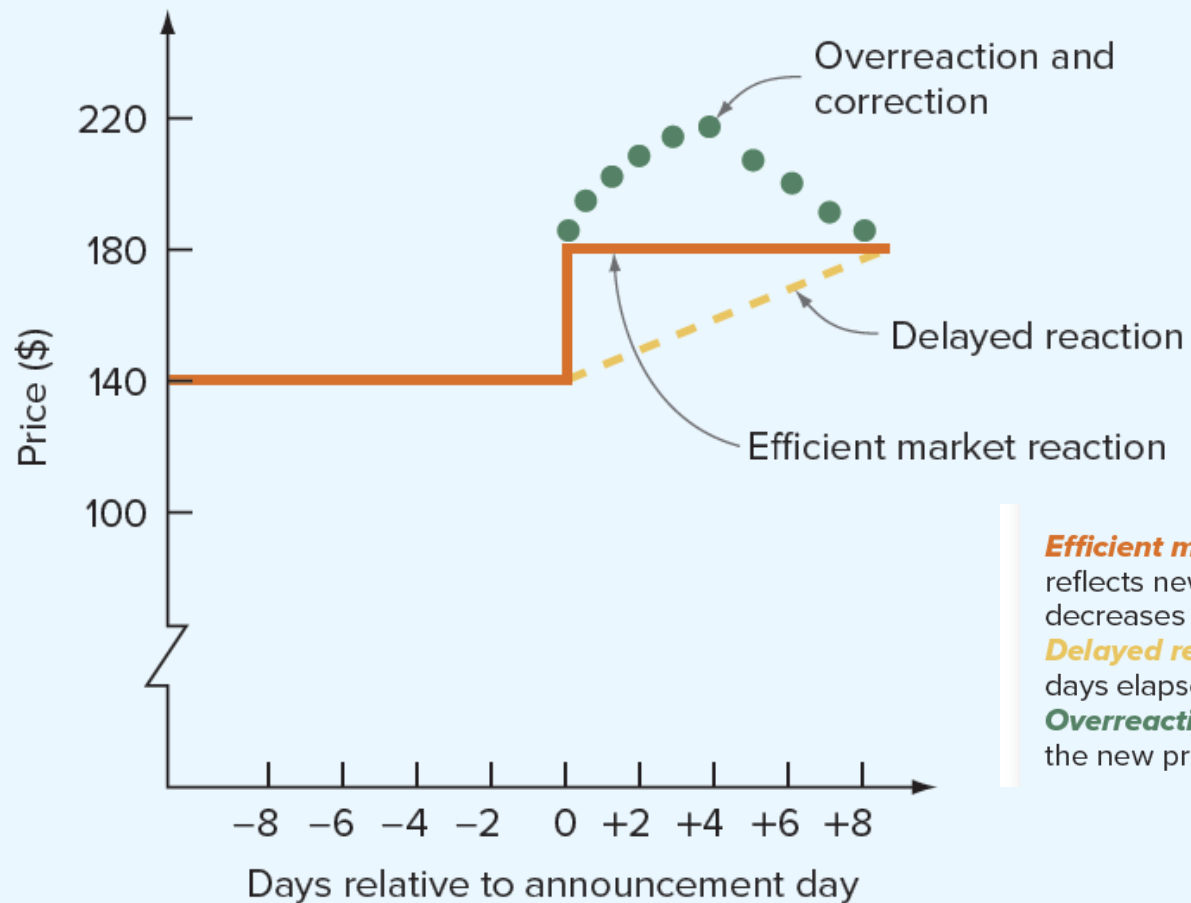
Series	Average Return		Standard Deviation
	Geometric	Arithmetic	
Large-company stocks	10.0%	12.0%	19.9%
Small-company stocks	12.0	16.6	31.9
Long-term corporate bonds	6.0	6.3	8.4
Long-term government bonds	5.5	6.0	9.9
Intermediate-term government bonds	5.3	5.3	5.6
U.S. Treasury bills	3.4	3.4	3.1
Inflation	2.9	3.0	4.1

SOURCE: Morningstar, 2017, author calculations.

EFFICIENT MARKETS HYPOTHESIS

- If the market is strong form efficient, **then all information of every kind is reflected in stock prices**. In such a market, there is no such thing as inside information.
- The second form of efficiency, semistrong form efficiency, is the most **controversial**. If a market is semistrong form efficient, then all public information is reflected in the stock price.
- The third form of efficiency, weak form efficiency, suggests that, at a **minimum**, the current price of a stock **reflects the stock's own past prices**.

MARKET REACTIONS



Efficient market reaction: The price instantaneously adjusts to and fully reflects new information; there is no tendency for subsequent increases and decreases to occur.

Delayed reaction: The price partially adjusts to the new information; eight days elapse before the price completely reflects the new information.

Overreaction: The price overadjusts to the new information; it overshoots the new price and subsequently corrects.

EXPECTED RETURNS AND VARIANCES

State of Economy	Probability of State of Economy	Rate of Return If State Occurs	
		Stock L	Stock U
Recession	.50	-20%	30%
Boom	.50	70	10
	1.00		

States of the Economy and Stock Returns

$$E(R) = \sum_{i=1}^n p_i R_i$$

$$E(R_U) = .50 \times 30\% + .50 \times 10\% = 20\%$$

$$E(R_L) = .50 \times -20\% + .50 \times 70\% = 25\%$$

(1) State of Economy	(2) Probability of State of Economy	Stock L		Stock U	
		(3) Rate of Return If State Occurs	(4) Product (2) × (3)	(5) Rate of Return If State Occurs	(6) Product (2) × (5)
Recession	.50	-.20	-.10	.30	.15
Boom	.50	.70	.35	.10	.05
	1.00	$E(R_L) = .25$, or 25%		$E(R_U) = .20$, or 20%	

RISK PREMIUM

- *Risk premium* as the difference between the expected return on a risky investment and the certain return on a risk-free investment.

Risk premium = Expected return – Risk-free rate

$$= E(R_U) - R_f$$

$$= 20\% - 8$$

$$= 12\%$$

risk premium on Stock L is $25\% - 8 = 17\%$.

RISK MEASURES

$$\sigma^2 = \left[\frac{\sum_{i=1}^n (R_i - \bar{R})^2}{n-1} \right]$$
$$\sigma = \sqrt{\sigma^2}$$

$$\sigma^2 = \sum p_i (R_i - E(R))^2$$

$$\sigma = (\sigma^2)^{1/2}$$

$$\text{Variance} = \sigma^2 = .50 \times .10^2 + .50 \times (-.10)^2 = .01$$

The standard deviation is the square root of this:

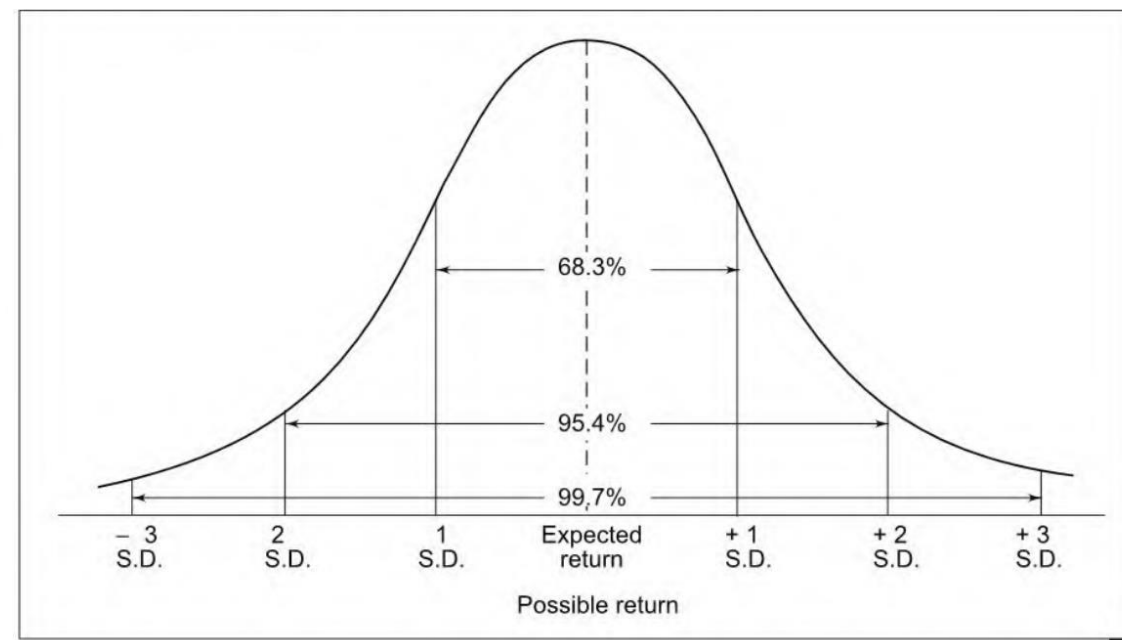
$$\text{Standard deviation} = \sigma = \sqrt{.01} = .10, \text{ or } 10\%$$

EXAMPLE

Bharat Foods Stock						
State of the Economy	p_i	R_i	$p_i R_i$	$R_i - E(R)$	$(R_i - E(R))^2$	$p_i (R_i - E(R))^2$
1. Boom	0.30	16	4.8	4.5	20.25	6.075
2. Normal	0.50	11	5.5	-0.5	0.25	0.125
3. Recession	0.20	6	1.2	-5.5	30.25	6.050
$\Sigma p_i R_i = 11.5$					$\Sigma p_i (R_i - E(R))^2 = 12.25$	
$\sigma = [\Sigma p_i (R_i - E(R))^2]^{1/2} = (12.25)^{1/2} = 3.5\%$						
Oriental Shipping Stock						
1. Boom	0.30	40	12.0	27.0	729.0	218.7
2. Normal	0.50	10	5.0	-3.0	0.25	4.5
3. Recession	0.20	-20	-4.0	-33.0	1089.00	217.8
$\Sigma p_i R_i = 13.0$					$\Sigma p_i (R_i - E(R))^2 = 441.0$	
$\sigma = [\Sigma p_i (R_i - E(R))^2]^{1/2} = (441.0)^{1/2} = 21.0\%$						

RETURN AND NORMAL DISTRIBUTION

<i>Band</i>	<i>Probability</i>
± One standard deviations from the expected return	68.3 %
± Two standard deviations from the expected return	95.4 %
± Three standard deviations from the expected return	99.7 %



Article: R. Cont (2001), “[Empirical properties of asset returns: stylized facts and statistical issues](https://doi.org/10.1080/713665670)” , 1:2, 223-236,
<https://doi.org/10.1080/713665670>

RISK AVERSION

- The relationship of a person's certainty equivalent (a benchmark return) to the expected monetary value of a risky investment defines his attitude toward risk.
- If the certainty equivalent is less than the expected value, the person is risk-averse;
- if the certainty equivalent is equal to the expected value, the person is risk-neutral;
- finally, if the certainty equivalent is more than the expected value, the person is risk-loving.

PORTFOLIO

- A group of assets such as stocks and bonds held by an investor. The percentage of a portfolio's total value that is invested in a particular asset.
- **Expected Return on a Portfolio**

$$E(R_p) = w_1 E(R_1) + (1 - w_1) E(R_2)$$

$$E(R_p) = \sum w_i E(R_i) \qquad E(R_p) = x_1 \times E(R_1) + x_2 \times E(R_2) + \cdots + x_n \times E(R_n)$$

EXAMPLE

$E(R_1) = 10$ percent, $E(R_2) = 12$ percent, $E(R_3) = 15$ percent, $E(R_4) = 18$ percent, $E(R_5) = 20$ percent. The portfolio proportions invested in these securities are: $w_1 = 0.1$, $w_2 = 0.2$, $w_3 = 0.3$, $w_4 = 0.2$, $w_5 = 0.2$. The expected portfolio return is:

$$\begin{aligned} E(R_p) &= w_1 E(R_1) + w_2 E(R_2) + w_3 E(R_3) + w_4 E(R_4) + w_5 E(R_5) \\ &= 0.1 \times 10 + 0.2 \times 12 + 0.3 \times 15 + 0.2 \times 18 + 0.2 \times 20 \\ &= 15.5 \text{ percent} \end{aligned}$$

EXAMPLE

<i>State of the Economy</i>	<i>Probability</i>	<i>Return on Stock A(%)</i>	<i>Return on Stock B(%)</i>	<i>Return on Portfolio(%)</i>
1	0.20	15	-5	5
2	0.20	-5	15	5
3	0.20	5	25	15
4	0.20	35	5	20
5	0.20	25	35	30

SOLUTION

<i>Expected Return</i>	
Stock A	: $0.2(15\%) + 0.2(-5\%) + 0.2(5\%) + 0.2(35\%) + 0.2(25\%) = 15\%$
Stock B	: $0.2(-5\%) + 0.2(15\%) + 0.2(25\%) + 0.2(5\%) + 0.2(35\%) = 15\%$
Portfolio of A and B	: $0.2(5\%) + 0.2(5\%) + 0.2(15\%) + 0.2(20\%) + 0.2(30\%) = 15\%$

RISK

Standard Deviation

Stock A	:	σ_A^2	=	$0.2(15-15)^2 + 0.2(-5-15)^2 + 0.2(5-15)^2 + 0.2(35-15)^2 + 0.2(25-15)^2$
			=	200
		σ_A	=	$(200)^{1/2} = 14.14\%$
Stock B	:	σ_B^2	=	$0.2(-5-15)^2 + 0.2(15-15)^2 + 0.2(25-15)^2 + 0.2(5-15)^2 + 0.2(35-15)^2$
			=	200
		σ_B	=	$(200)^{1/2} = 14.14\%$
Portfolio	:	$\sigma_{(A+B)}^2$	=	$0.2(5-15)^2 + 0.2(5-15)^2 + 0.2(15-15)^2 + 0.2(20-15)^2 + 0.2(30-15)^2$
			=	90
		σ_{A+B}^2	=	$(90)^{1/2} = 9.49\%$

PROBLEM FOR STUDENTS

- Suppose a portfolio, which has 50 percent in Stock A and 25 percent in each of Stocks B and C. what is the standard deviations of this portfolio?

State of Economy	Probability of State of Economy	Rate of Return If State Occurs		
		Stock A	Stock B	Stock C
Boom	.40	10%	15%	20%
Bust	.60	8	4	0

SOLUTION

State of Economy	Probability of State of Economy	Rate of Return If State Occurs			
		Stock A	Stock B	Stock C	Portfolio
Boom	.40	10%	15%	20%	13.75%
Bust	.60	8	4	0	5.00

The portfolio return when the economy booms is calculated as:

$$E(R_p) = .50 \times 10\% + .25 \times 15\% + .25 \times 20\% = 13.75\%$$

The return when the economy goes bust is calculated the same way. The expected return on the portfolio is 8.5 percent. The variance is thus:

$$\begin{aligned}\sigma_p^2 &= .40 \times (.1375 - .085)^2 + .60 \times (.05 - .085)^2 \\ &= .0018375\end{aligned}$$

The standard deviation is about 4.3 percent. For our equally weighted portfolio, check to see that the standard deviation is about 5.4 percent.

$$E(R_A) = 8.8\%$$

$$E(R_B) = 8.4\%$$

$$E(R_C) = 8.0\%$$

Weight	Return (%)
0.5	8.80%
0.25	8.40%
0.25	8.00%
	8.50%

PROBLEM: RISK AND RETURN

- Consider the following information:

<u>State</u>	<u>Probability</u>	<u>ABC, Inc. Return</u>
Boom	.25	0.15
Normal	.50	0.08
Slowdown	.15	0.04
Recession	.10	-0.03

- What is the expected return?
- What is the variance?
- What is the standard deviation?

SOLUTION

- $E(R) = .25(0.15) + .5(0.08) + .15(0.04) + .1(-0.03) = 8.05\%$
- $\text{Variance} = .25(.15-0.0805)^2 + .5(0.08-0.0805)^2 + .15(0.04-0.0805)^2 + .1(-0.03-0.0805)^2 = 0.00267475$
- $\text{Standard Deviation} = 5.17\%$

PORTFOLIOS

- A portfolio is a collection of assets.
- An asset's risk and return are important in how they affect the risk and return of the portfolio.
- The risk-return trade-off for a portfolio is measured by the portfolio expected return and standard deviation, just as with individual assets.

EXAMPLE: PORTFOLIO WEIGHTS

- Suppose you have \$15,000 to invest and you have purchased securities in the following amounts. What are your portfolio weights in each security?

- \$2000 of C
- \$3000 of KO
- \$4000 of INTC
- \$6000 of BP

- C: $2/15 = .133$
- KO: $3/15 = .2$
- INTC: $4/15 = .267$
- BP: $6/15 = .4$

the sum of the weights = 1

PORTFOLIO EXPECTED RETURNS

- The expected return of a portfolio is the weighted average of the expected returns of the respective assets in the portfolio.

$$E(R_P) = \sum_{j=1}^m w_j E(R_j)$$

- You can also find the expected return by finding the portfolio return in each possible state and computing the expected value as we did with individual securities.

EXAMPLE: EXPECTED PORTFOLIO RETURNS

- Consider the portfolio weights computed previously. If the individual stocks have the following expected returns, what is the expected return for the portfolio?
 - C: 19.69%
 - KO: 5.25%
 - INTC: 16.65%
 - BP: 18.24%
- $E(R_P) = .133(19.69\%) + .2(5.25\%) + .267(16.65\%) + .4(18.24\%) = 15.41\%$

PORTFOLIO VARIANCE

- Compute the portfolio return for each state:

$$R_P = w_1R_1 + w_2R_2 + \dots + w_mR_m$$

- Compute the expected portfolio return using the same formula as for an individual asset.
- Compute the portfolio variance and standard deviation using the same formulas as for an individual asset.

EXAMPLE: PORTFOLIO VARIANCE

- Consider the following information on returns and probabilities:
 - Invest 50% of your money in Asset A.

State	Probability	A	B	Portfolio
Boom	.4	30%	-5%	12.5%
Bust	.6	-10%	25%	7.5%

- What are the expected return and standard deviation for each asset?
- What are the expected return and standard deviation for the portfolio?

SOLUTION

If A and B are your only choices, what percent are you investing in Asset B? 50%

- Asset A: $E(R_A) = .4(30) + .6(-10) = 6\%$

$$\text{Variance}(A) = .4(30-6)^2 + .6(-10-6)^2 = 384$$

$$\text{Std. Dev.}(A) = 19.6\%$$

- Asset B: $E(R_B) = .4(-5) + .6(25) = 13\%$

$$\text{Variance}(B) = .4(-5-13)^2 + .6(25-13)^2 = 216$$

$$\text{Std. Dev.}(B) = 14.7\%$$

- Portfolio :-

$$\text{Portfolio return in boom} = .5(30) + .5(-5) = 12.5$$

$$\text{Portfolio return in bust} = .5(-10) + .5(25) = 7.5$$

SOLUTION - CONTINUE....

- Expected return = $.4(12.5) + .6(7.5) = 9.5$ or
Expected return = $.5(6) + .5(13) = 9.5$
Variance of portfolio = $.4(12.5-9.5)^2 + .6(7.5-9.5)^2 = 6$
Standard deviation = 2.45%
- What would the expected return and standard deviation for the portfolio be if we invested 3/7 of our money in A and 4/7 in B? Portfolio return = 10% and standard deviation = 0
- Portfolio variance using covariances:
 $COV(A,B) = .4(30-6)(-5-13) + .6(-10-6)(25-13) = -288$
Variance of portfolio = $(.5)^2(384) + (.5)^2(216) + 2(.5)(.5)(-288) = 6$
Standard deviation = 2.45%

PORTFOLIO RISK: TWO ASSETS

$$\text{Portfolio variance} = x_1^2 \sigma_1^2 + x_2^2 \sigma_2^2 + 2(x_1 x_2 \rho_{12} \sigma_1 \sigma_2)$$

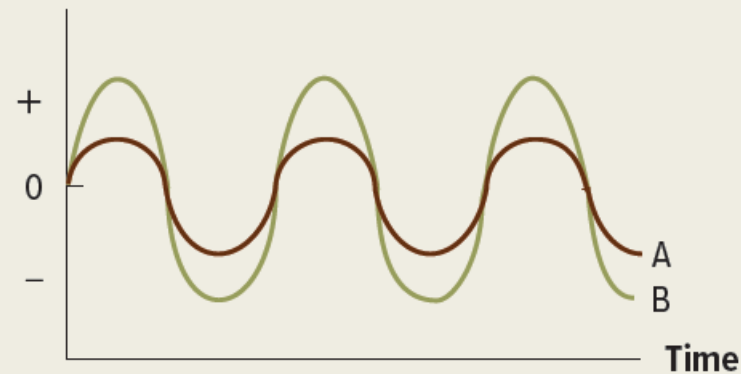
	Stock 1	Stock 2
Stock 1	$x_1^2 \sigma_1^2$	$x_1 x_2 \sigma_{12}$ $= x_1 x_2 \rho_{12} \sigma_1 \sigma_2$
Stock 2	$x_1 x_2 \sigma_{12}$ $= x_1 x_2 \rho_{12} \sigma_1 \sigma_2$	$x_2^2 \sigma_2^2$

CORRELATION

Perfect positive correlation

$$\text{Corr}(R_A, R_B) = 1$$

Returns



Both the return on Security A and the return on Security B are higher than average at the same time. Both the return on Security A and the return on Security B are lower than average at the same time.

Perfect negative correlation

$$\text{Corr}(R_A, R_B) = -1$$

Returns

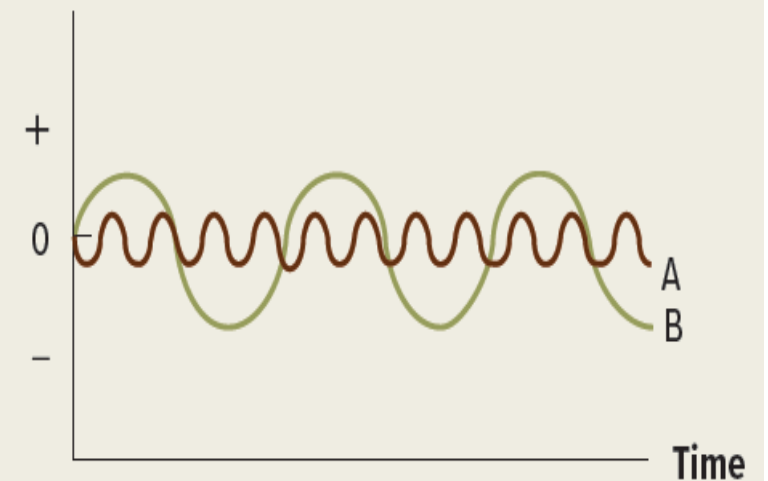


Security A has a higher-than-average return when Security B has a lower-than-average return, and vice versa.

Zero correlation

$$\text{Corr}(R_A, R_B) = 0$$

Returns



The return on Security A is completely unrelated to the return on Security B.

PORTFOLIO THEORY- OPPORTUNITY SET

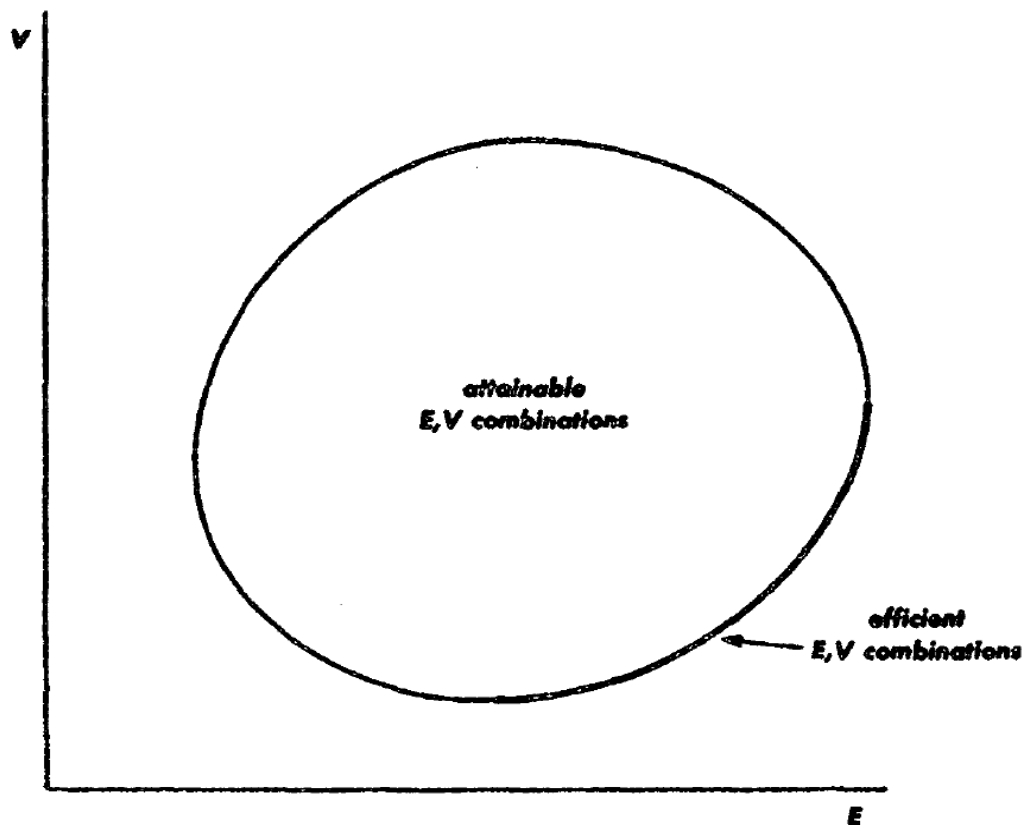


FIG. 1

The *E-V rule states* that the investor would (or should) want to select one of those portfolios which give rise to the (E, V) combinations indicated as efficient in the figure; i.e., those with minimum V for given E or more and maximum E for given V or less.

Portfolio Selection

Harry Markowitz

The Journal of Finance, Vol. 7, No. 1 (Mar., 1952), pp. 77-91 (15 pages)

<https://doi.org/10.2307/2975974>•<https://www.jstor.org/stable/2975974>

EXAMPLE - BOND AND STOCK PORTFOLIO

The bond index is an index of corporate bond returns. The historical data are

$$\bar{R}_S = 11.8\% \quad \sigma_S = 20.3\% \quad \rho_{SB} = .01$$

$$\bar{R}_B = 6.4\% \quad \sigma_B = 8.4\%$$

Table 4.10 Mean Return and Standard Deviation for Combinations of Stocks and Bonds

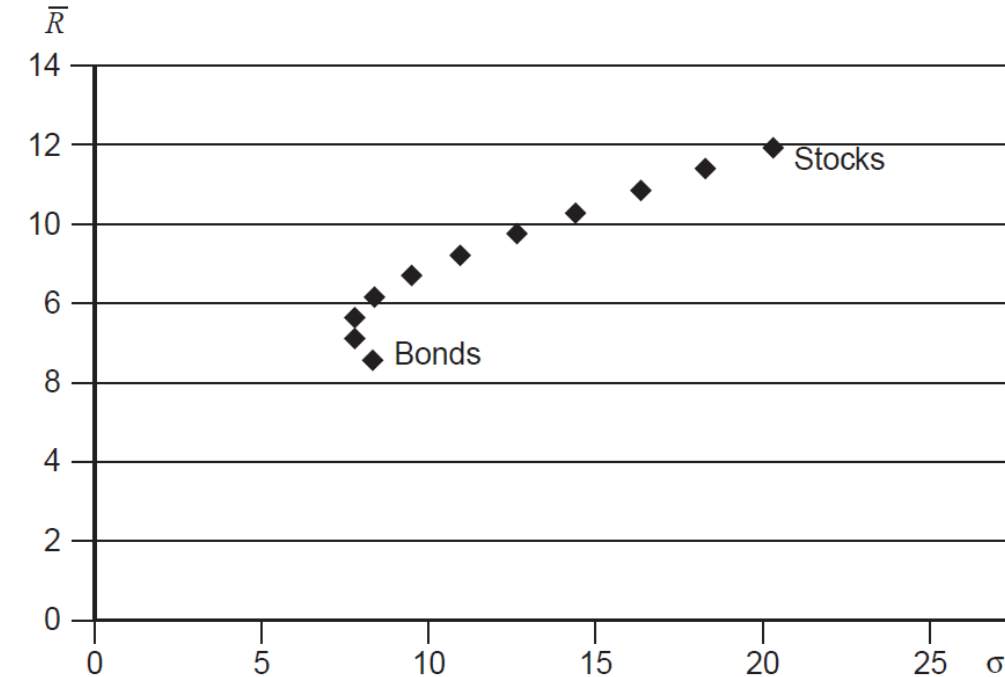
Proportion Stocks	Proportion Bonds	Mean Return	Standard Deviation
1	0	11.8	20.3
0.9	0.1	11.26	18.29
0.8	0.2	10.72	16.33
0.7	0.3	10.18	14.43
0.6	0.4	9.64	12.63
0.5	0.5	9.1	10.98
0.4	0.6	8.56	9.56
0.3	0.7	8.02	8.47
0.2	0.8	7.48	7.85
0.1	0.9	6.94	7.83
0	1	6.4	8.40

OPPORTUNITY SET

The bond index is an index of corporate bond returns. The historical data are

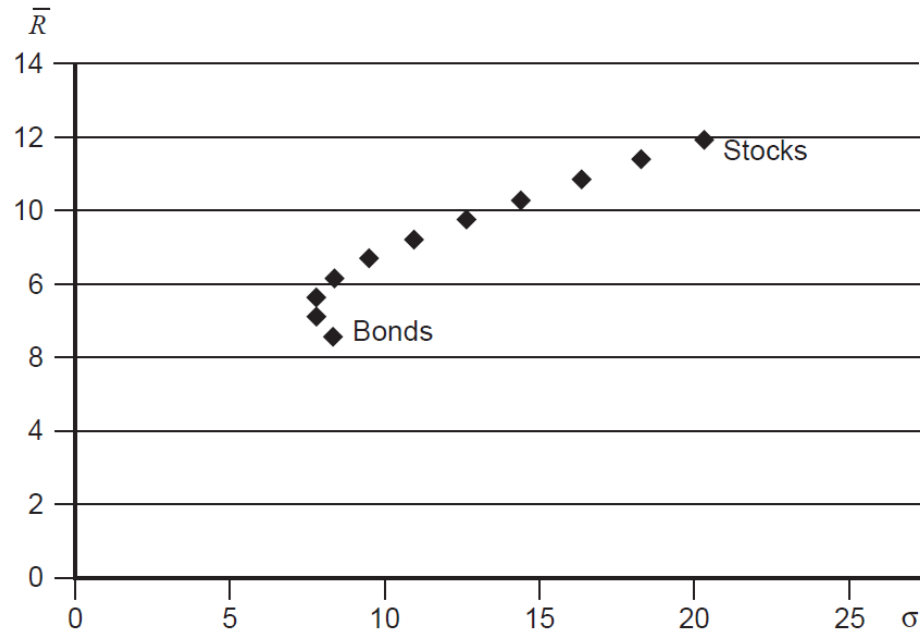
$$\bar{R}_S = 11.8\% \quad \sigma_S = 20.3\% \quad \rho_{SB} = .01$$

$$\bar{R}_B = 6.4\% \quad \sigma_B = 8.4\%$$

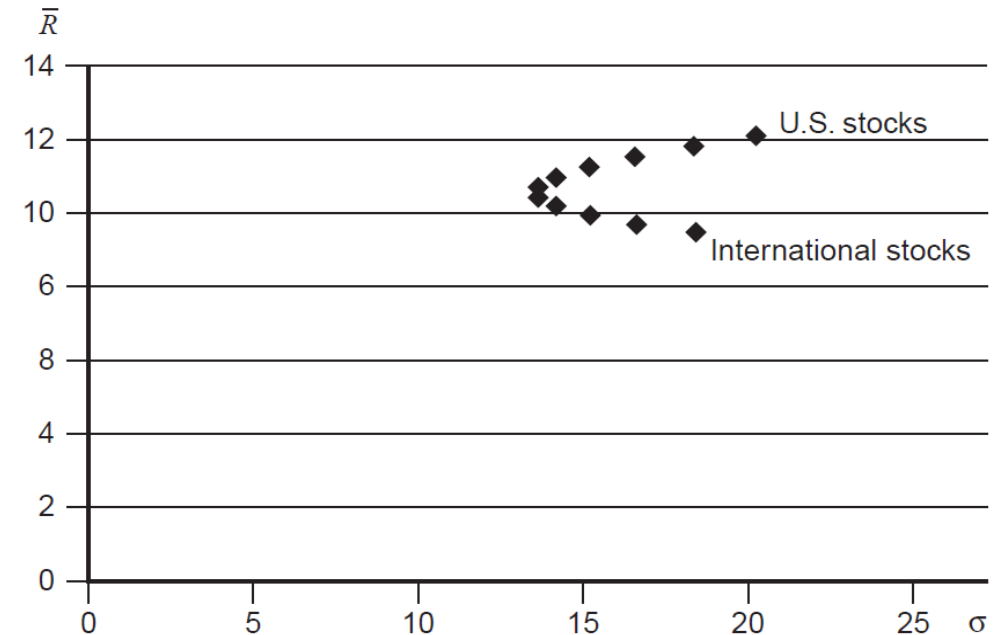


Combinations of bonds and stocks.

TWO ASSET EXAMPLE



Combinations of bonds and stocks.



Combinations of U.S. stocks and international stocks.

CHARACTERISTICS OF PORTFOLIOS

If R_{Pj} is the j th return on the portfolio and X_i is the fraction of the investor's funds invested in the i th asset, and N is the number of assets, then

$$R_{Pj} = \sum_{i=1}^N (X_i R_{ij})$$

Taking the expected value of the expression just given for the return on a portfolio yields

$$\bar{R}_P = E(R_P) = E\left(\sum_{i=1}^N X_i R_{ij}\right)$$

$$\bar{R}_P = \sum_{i=1}^N E(X_i R_{ij}) = \sum_{i=1}^N (X_i \bar{R}_i)$$

A MEASURE OF DISPERSION

$$\sigma_P^2 = X_1^2 \sigma_1^2 + X_2^2 \sigma_2^2 + 2X_1 X_2 \sigma_{12}$$

$$\sigma_P^2 = \sum_{j=1}^N (X_j^2 \sigma_j^2) + \sum_{j=1}^N \sum_{\substack{k=1 \\ k \neq j}}^N (X_j X_k \sigma_{jk})$$

$$\sigma_P^2 = X_1^2 \sigma_1^2 + X_2^2 \sigma_2^2 + X_3^2 \sigma_3^2 + 2X_1 X_2 \sigma_{12} + 2X_1 X_3 \sigma_{13} + 2X_2 X_3 \sigma_{23}$$

VARIANCE OF PORTFOLIO YIELD (INDEPENDENT ASSETS)

$$\sigma_P^2 = \sum_{j=1}^N (X_j^2 \sigma_j^2) + \sum_{j=1}^N \sum_{\substack{k=1 \\ k \neq j}}^N (X_j X_k \sigma_{jk})$$

the case where all assets are independent and therefore the covariance between them is zero. In this case, $\sigma_{jk} = 0$,


$$\sum_{j=1}^N (X_j^2 \sigma_j^2)$$

Furthermore, assume equal amounts are invested in each asset. With N assets, the proportion invested in each asset is $1/N$. Applying our formula yields

$$\sigma_P^2 = \sum_{j=1}^N (1/N)^2 \sigma_j^2 = 1/N \left[\sum_{j=1}^N \frac{\sigma_j^2}{N} \right]$$

As N becomes extremely large, the variance of the **portfolio approaches zero**. This is a general result. If assets in portfolio are *independent*, the variance of a portfolio of these assets approaches zero.

VARIANCE OF PORTFOLIO YIELD

$$\sigma_P^2 = \sum_{j=1}^N (X_j^2 \sigma_j^2) + \sum_{j=1}^N \sum_{\substack{k=1 \\ k \neq j}}^N (X_j X_k \sigma_{jk})$$

With equal investment, the proportion invested in any one asset X_j is $1/N$, and the formula for the variance of a portfolio becomes

$$\sigma_P^2 = \sum_{j=1}^N (1/N)^2 \sigma_j^2 + \sum_{j=1}^N \sum_{\substack{k=1 \\ k \neq j}}^N (1/N)(1/N) \sigma_{jk}$$

Factoring out $1/N$ from the first summation and $(N-1)/N$ from the second yields

$$\sigma_P^2 = (1/N) \sum_{j=1}^N \left[\frac{\sigma_j^2}{N} \right] + \frac{(N-1)}{N} \sum_{j=1}^N \sum_{\substack{k=1 \\ k \neq j}}^N \left[\frac{\sigma_{jk}}{N(N-1)} \right]$$

there are $N(N-1)$ covariance terms. Thus, the second term is the summation of covariances divided by the number of covariances, and it is, therefore, an average.

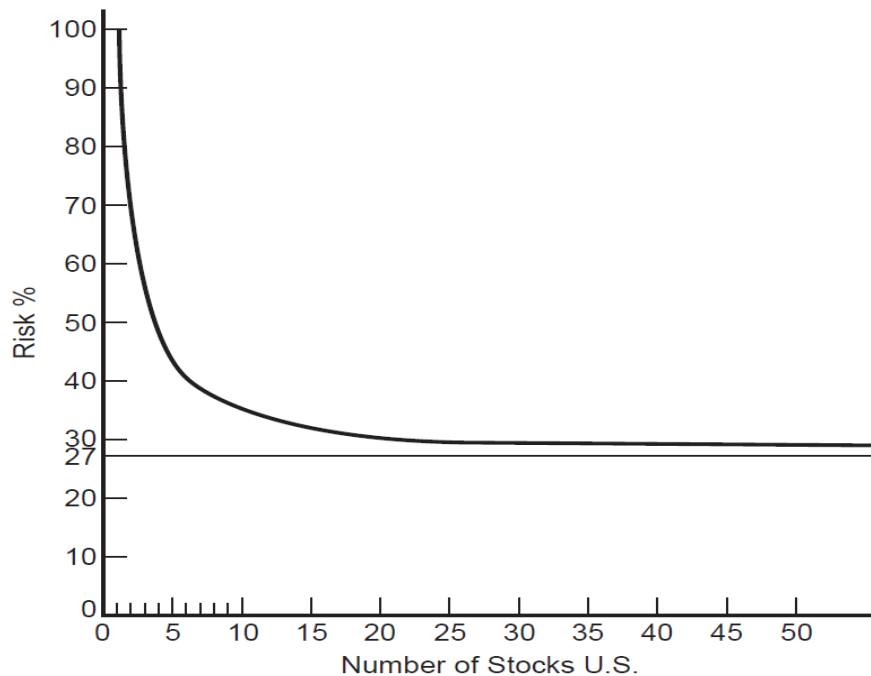
$$\sigma_P^2 = \frac{1}{N} \bar{\sigma}_j^2 + \frac{N-1}{N} \bar{\sigma}_{jk}$$

EFFECT OF DIVERSIFICATION ON PORTFOLIO

$$\sigma_P^2 = \frac{1}{N} \bar{\sigma}_j^2 + \frac{N-1}{N} \bar{\sigma}_{jk}$$



The contribution to the portfolio variance of the variance of the individual securities goes to zero as N gets very large. However, the contribution of the covariance terms approaches the average covariance as N gets large.



Percentage of the Risk on an Individual Security That Can Be Eliminated by Holding a Random Portfolio of Stocks within Selected National Markets and among National Markets (1975)

United States	73
United Kingdom	65.5
France	67.3
Germany	56.2
Italy	60.0
Belgium	80.0
Switzerland	56.0
Netherlands	76.1
International stocks	89.3

PROBLEM FOR STUDENTS

- Consider the following information on returns and probabilities:

<u>State</u>	<u>Probability</u>	<u>X</u>	<u>Z</u>
Boom	.25	15%	10%
Normal	.60	10%	9%
Recession	.15	5%	10%

- What are the expected return and standard deviation for a portfolio with an investment of \$6,000 in asset X and \$4,000 in asset Z?

SOLUTION

- Portfolio return in Boom: $.6(15) + .4(10) = 13\%$
- Portfolio return in Normal: $.6(10) + .4(9) = 9.6\%$
- Portfolio return in Recession: $.6(5) + .4(10) = 7\%$
 - Expected return = $.25(13) + .6(9.6) + .15(7) = 10.06\%$
 - Variance = $.25(13-10.06)^2 + .6(9.6-10.06)^2 + .15(7-10.06)^2 = 3.6924$
 - Standard deviation = 1.92%
- Compare to return on X of 10.5% and standard deviation of 3.12%
- And return on Z of 9.4% and standard deviation of $.49\%$
- Using covariances:
 - $\text{COV}(X,Z) = .25(15-10.5)(10-9.4) + .6(10-10.5)(9-9.4) + .15(5-10.5)(10-9.4) = .3$
 - Portfolio variance = $(.6 \times 3.12)^2 + (.4 \times .49)^2 + 2(.6)(.4)(.3) = 3.6868$
 - Portfolio standard deviation = 1.92% (difference in variance due to rounding)

EXPECTED VS. UNEXPECTED RETURNS

- Realized returns are generally not equal to expected returns.
- There is the expected component and the unexpected component.
 - At any point in time, the unexpected return can be either positive or negative.
 - Over time, the average of the unexpected component is zero.

RETURN COMPONENT

- EXPECTED AND UNEXPECTED RETURNS

Total return = Expected return + Unexpected return

$$R = E(R) + U$$

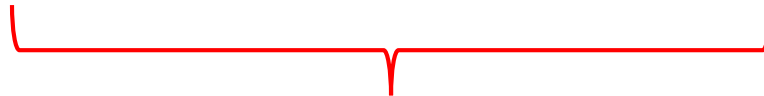
return from the stock is the part of the return that shareholders in the market predict or expect.

return on the stock is the uncertain, or risky, part. (unexpected information within the year).

RISK: SYSTEMATIC AND UNSYSTEMATIC

- ***systematic risk***: A risk that influences a large number of assets. Also called market risk.
- ***unsystematic risk***: A risk that affects at most a small number of assets. Also called unique or asset-specific risk.

$$R = E(R) + \text{Systematic portion} + \text{Unsystematic portion}$$



Unexpected return

SYSTEMATIC RISK

- Risk factors that affect a large number of assets
- Also known as non-diversifiable risk or market risk
- Includes such things as changes in GDP, inflation, interest rates, etc.

UNSYSTEMATIC RISK

- Risk factors that affect a limited number of assets
- Also known as unique risk and asset-specific risk
- Includes such things as labor strikes, part shortages, etc.

RISKY ASSETS PORTFOLIO VARIANCE

$$\begin{aligned}\text{Portfolio variance} &= N \left(\frac{1}{N} \right)^2 \times \text{average variance} \\ &\quad + (N^2 - N) \left(\frac{1}{N} \right)^2 \times \text{average covariance} \\ &= \frac{1}{N} \times \text{average variance} + \left(1 - \frac{1}{N} \right) \times \text{average covariance}\end{aligned}$$

Notice that as N increases, the portfolio variance steadily approaches the average covariance. If the average covariance were zero, it would be possible to eliminate all risk by holding a sufficient number of securities.

RISK COMPONENT

$$\text{Total risk} = \text{Systematic risk} + \text{Unsystematic risk}$$

Systematic risk is also called nondiversifiable risk or market risk. Unsystematic risk is also called diversifiable risk, unique risk, or asset-specific risk. For a well-diversified portfolio, the unsystematic risk is negligible. For such a portfolio, essentially all of the risk is systematic.

The systematic risk principle states that the reward for bearing risk depends only on the systematic risk of an investment. Because unsystematic risk can be eliminated at virtually no cost (by diversifying), there is no reward for bearing it. Put another way, the market does not reward risks that are borne unnecessarily.

The expected return on an asset depends only on that asset's systematic risk.

MEASURING SYSTEMATIC RISK

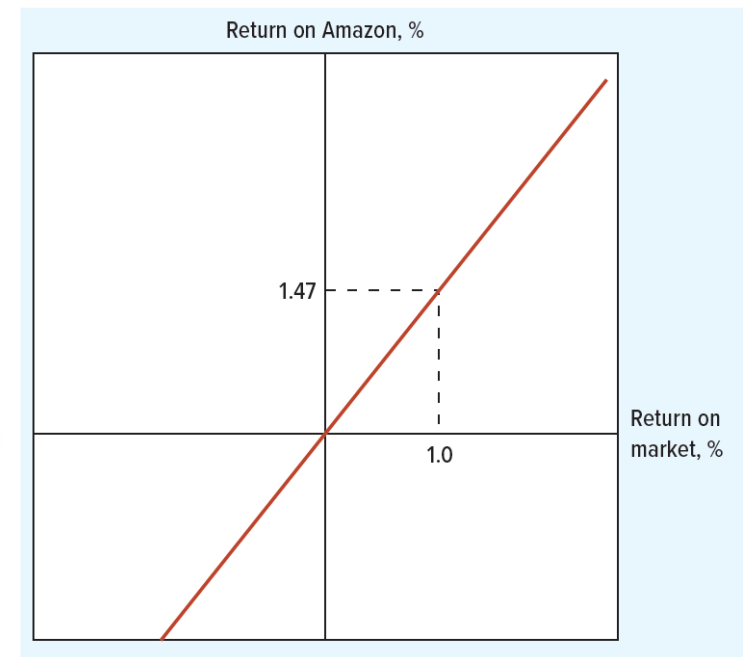
- MEASURING SYSTEMATIC RISK

- **beta coefficient** The amount of systematic risk present in a particular risky asset relative to that in an average asset.

$$\beta_i = \sigma_{im} / \sigma_m^2$$
$$\beta_j = \frac{\text{Cov}(R_j, R_M)}{\sigma_M^2} = \frac{\rho_{jM} \sigma_j \sigma_M}{\sigma_M^2} = \frac{\rho_{jM} \sigma_j}{\sigma_M}$$
$$\frac{\text{Cov}(R_j, R_M)}{\sigma_M^2} = \sum_{i=1}^n (R_{jt} - \bar{R}_j)(R_{Mt} - \bar{R}_M) / (n - 1)$$

- Portfolio Beta:

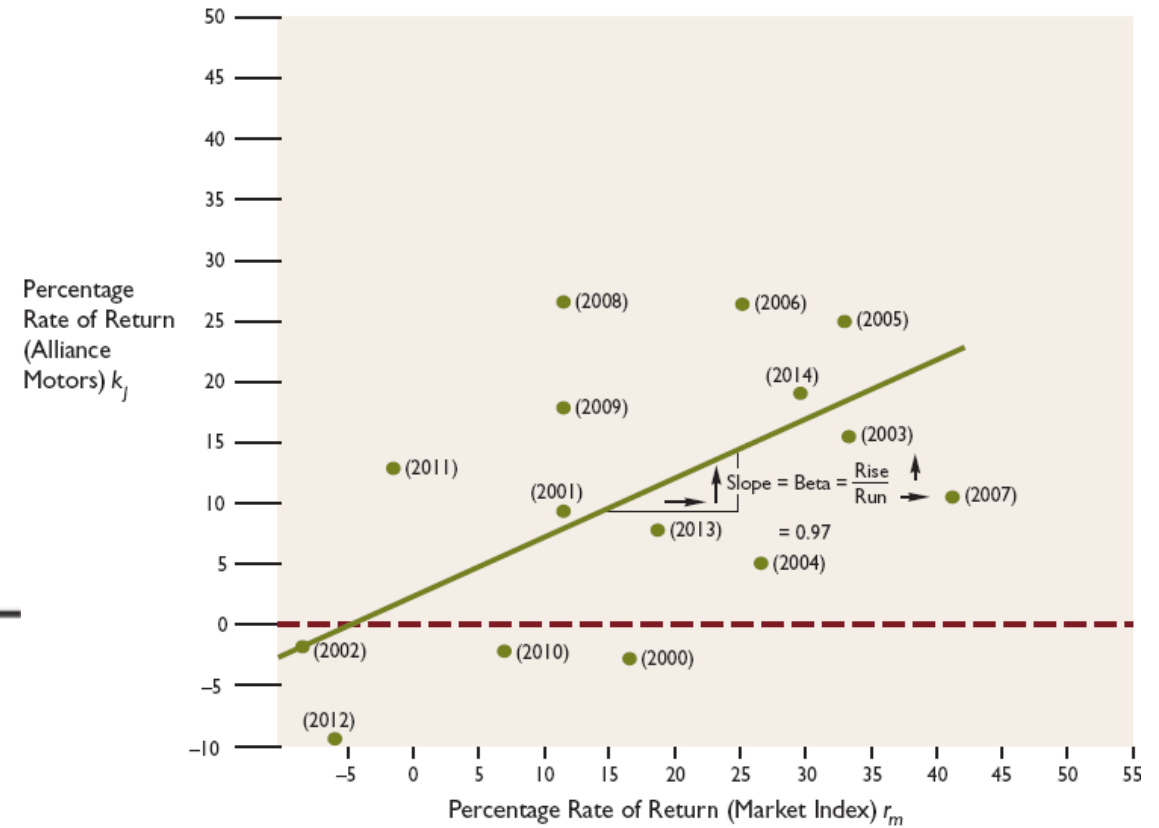
$$\beta_p = \sum_{i=1}^n w_i \beta_i$$



BETA THROUGH GRAPH

Interpretation of Selected Beta Coefficients

Beta Value	Direction of Movement in Returns	Interpretation
2.0	Same as market	Twice as risky (responsive) as market
1.0	Same as market	Risk equal to that of market
0.5	Same as market	Half as risky as market
0	Uncorrelated with market movements	No market-related risk
-0.5	Opposite of market	Half as responsive as market but in the opposite direction



BETA FOR SELECTED STOCKS

Company	Value Line Beta	Company	Value Line Beta
Amazon.com	1.05	Hershey Foods	0.65
American Electric Power	0.70	Honeywell	1.10
American States Water	0.70	IBM	0.85
Apple	0.95	Intel	1.00
Avis Budget group	1.50	Johnson & Johnson	0.75
Bank of America	1.45	JPMorgan Chase	1.20
Berkshire Hathaway	0.80	Kellogg	0.65
Boeing	1.05	Krispy Kreme	1.15
Cheesecake Factory	0.85	Microsoft	1.00
Citigroup	1.45	Nike	0.80
Coca-Cola	0.70	Southwest Airlines	1.05
Coca-Cola Bottling	0.65	Target	0.70
Duke Energy	0.65	United Continental Holdings	1.05
DuPont	1.10	Wal-Mart Stores	0.65
ExxonMobil	0.95	Whirlpool	1.30
General Electric	1.15	WEC Energy Group	0.70
General Mills	0.70	Wynn Resorts	1.35
Harley-Davidson	1.15	Yahoo! Inc.	1.05

Source: *The Value Line Investment Survey* (New York: Value Line Publishing, Inc., November 27, 2015). © 2015 by Value Line Publishing, Inc. Used by permission. All rights reserved.

EXAMPLE: PORTFOLIO BETA

Security	Amount Invested	Expected Return	Beta
Stock A	\$1,000	8%	.80
Stock B	2,000	12	.95
Stock C	3,000	15	1.10
Stock D	4,000	18	1.40

What is the expected return on this portfolio? What is the beta of this portfolio? Does this portfolio have more or less systematic risk than an average asset?

SOLUTION

$$\begin{aligned}E(R_p) &= .10 \times E(R_A) + .20 \times E(R_B) + .30 \times E(R_C) + .40 \times E(R_D) \\&= .10 \times 8\% + .20 \times 12\% + .30 \times 15\% + .40 \times 18\% \\&= 14.9\%\end{aligned}$$

Similarly, the portfolio beta, β_p , is:

$$\begin{aligned}\beta_p &= .10 \times \beta_A + .20 \times \beta_B + .30 \times \beta_C + .40 \times \beta_D \\&= .10 \times .80 + .20 \times .95 + .30 \times 1.10 + .40 \times 1.40 \\&= 1.16\end{aligned}$$

DETERMINANT OF BETA

- Cyclicalities of Revenues
- Operating Leverage
- Financial Leverage

TOTAL VS. SYSTEMATIC RISK

- Consider the following information:

	Standard Deviation	Beta
Security C	20%	1.25
Security K	30%	0.95

- Which security has more total risk?
- Which security has more systematic risk?
- Which security should have the higher expected return?

Security K has the higher total risk.

Security C has the higher systematic risk.

Security C should have the higher expected return.

PROBLEM: STUDENTS

- Consider the previous example with the following four securities.

<u>Security</u>	<u>Weight</u>	<u>Beta</u>
C	.133	1.685
KO	.2	0.195
INTC	.267	1.161
BP	.4	1.434

- What is the portfolio beta?

- $.133(1.685) + .2(.195) + .267(1.161) + .4(1.434) = 1.147$

Which security has the highest systematic risk? **C**

Which security has the lowest systematic risk? **KO**

Is the systematic risk of the portfolio more or less than the market? **more**

BETA AND RISK PREMIUM

Consider a portfolio made up of Asset A and a risk-free asset

Asset A has an expected return of $E(R_A) = 20\%$ and a beta of $\beta_A = 1.6$.
the risk-free rate is $R_f = 8\%$.

if 25 percent of the portfolio is invested in Asset A, then
expected return is:

$$\begin{aligned} E(R_p) &= .25 \times E(R_A) + (1 - .25) \times R_f \\ &= .25 \times 20\% + .75 \times 8\% \\ &= 11\% \end{aligned}$$

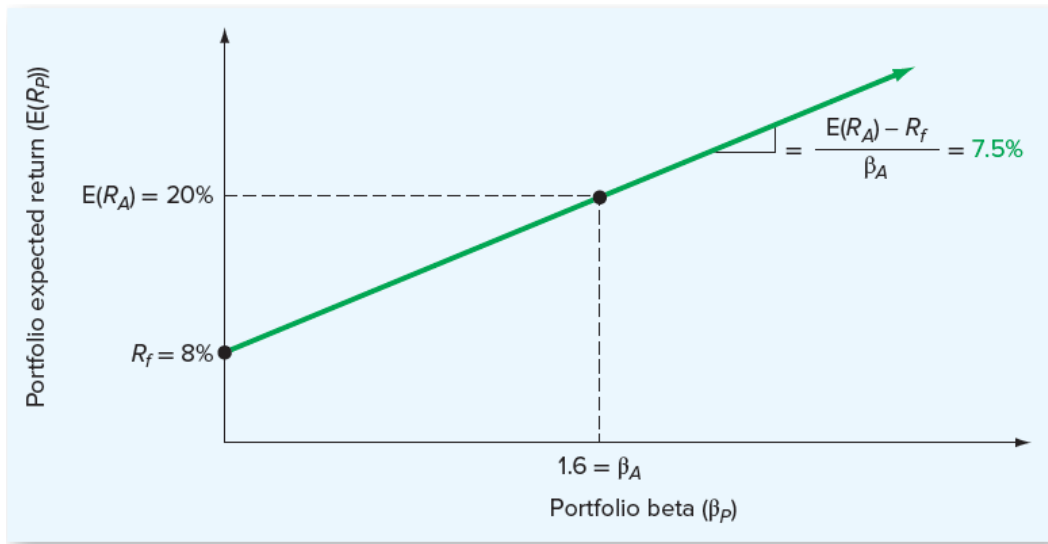
Similarly, the beta on the portfolio, β_p , would be:

$$\begin{aligned} \beta_p &= .25 \times \beta_A + (1 - .25) \times 0 \\ &= .25 \times 1.6 \\ &= .40 \end{aligned}$$

We can calculate some other possibilities, as follows:

Percentage of Portfolio in Asset A	Portfolio Expected Return	Portfolio Beta
0%	8%	.0
25	11	.4
50	14	.8
75	17	1.2
100	20	1.6
125	23	2.0
150	26	2.4

REWARD TO RISK RATIO



Portfolio Expected
Returns and Betas
for Asset A

$$\begin{aligned}\text{Slope} &= \frac{E(R_A) - R_f}{\beta_A} \\ &= \frac{.20 - .08}{1.6} = .075, \text{ or } 7.5\%\end{aligned}$$

, Asset A has a risk premium of 7.5 percent per “unit” of systematic risk.

EXTENSION - BETA & RISK PREMIUM

- Suppose we consider a second asset, Asset B. This asset has a beta of 1.2 and an expected return of 16 percent. Which investment is better, Asset A or Asset B?

$$\begin{aligned}E(R_p) &= .25 \times E(R_B) + (1 - .25) \times R_f \\&= .25 \times 16\% + .75 \times 8\% \\&= 10\%\end{aligned}$$

Similarly, the beta on the portfolio, β_p , will be:

$$\begin{aligned}\beta_p &= .25 \times \beta_B + (1 - .25) \times 0 \\&= .25 \times 1.2 \\&= .30\end{aligned}$$

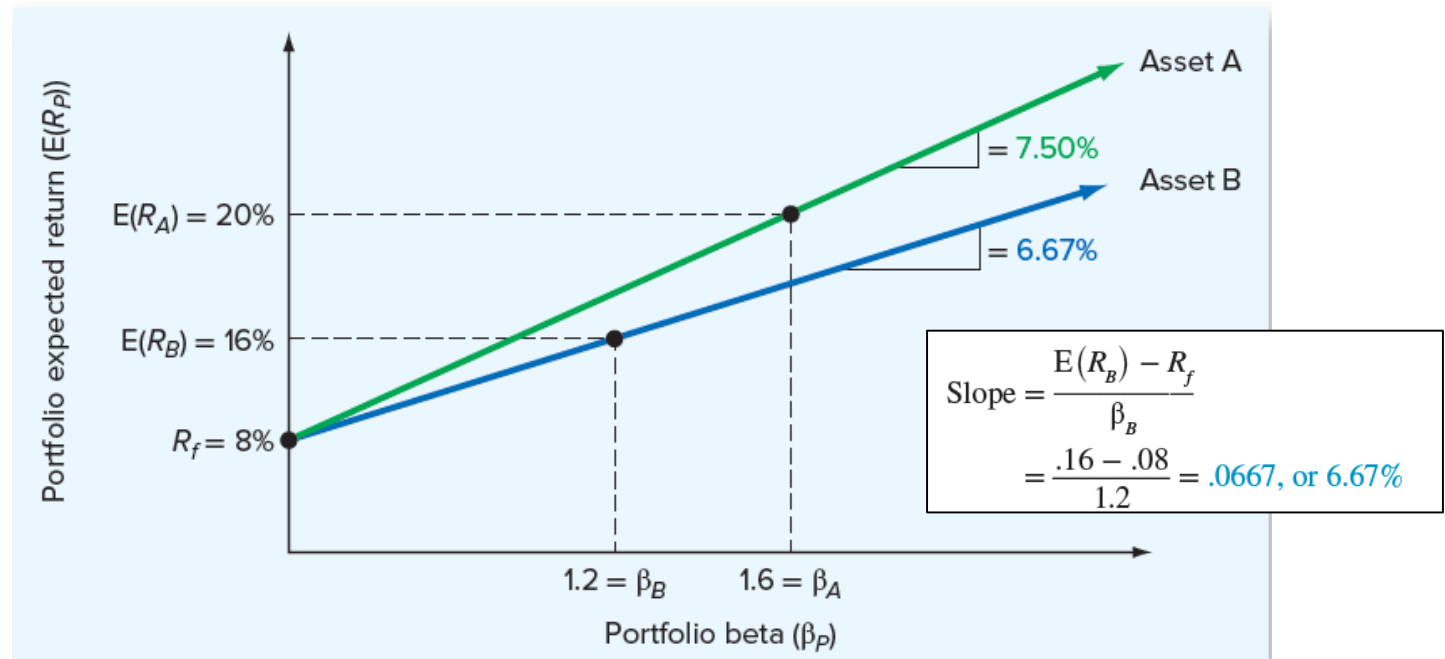
Some other possibilities are as follows:

Percentage of Portfolio in Asset B	Portfolio Expected Return	Portfolio Beta
0%	8%	.0
25	10	.3
50	12	.6
75	14	.9
100	16	1.2
125	18	1.5
150	20	1.8

SECURITY: COMPARE A AND B

Percentage of Portfolio in Asset A	Portfolio Expected Return	Portfolio Beta
0%	8%	.0
25	11	.4
50	14	.8
75	17	1.2
100	20	1.6
125	23	2.0
150	26	2.4

Percentage of Portfolio in Asset B	Portfolio Expected Return	Portfolio Beta
0%	8%	.0
25	10	.3
50	12	.6
75	14	.9
100	16	1.2
125	18	1.5
150	20	1.8

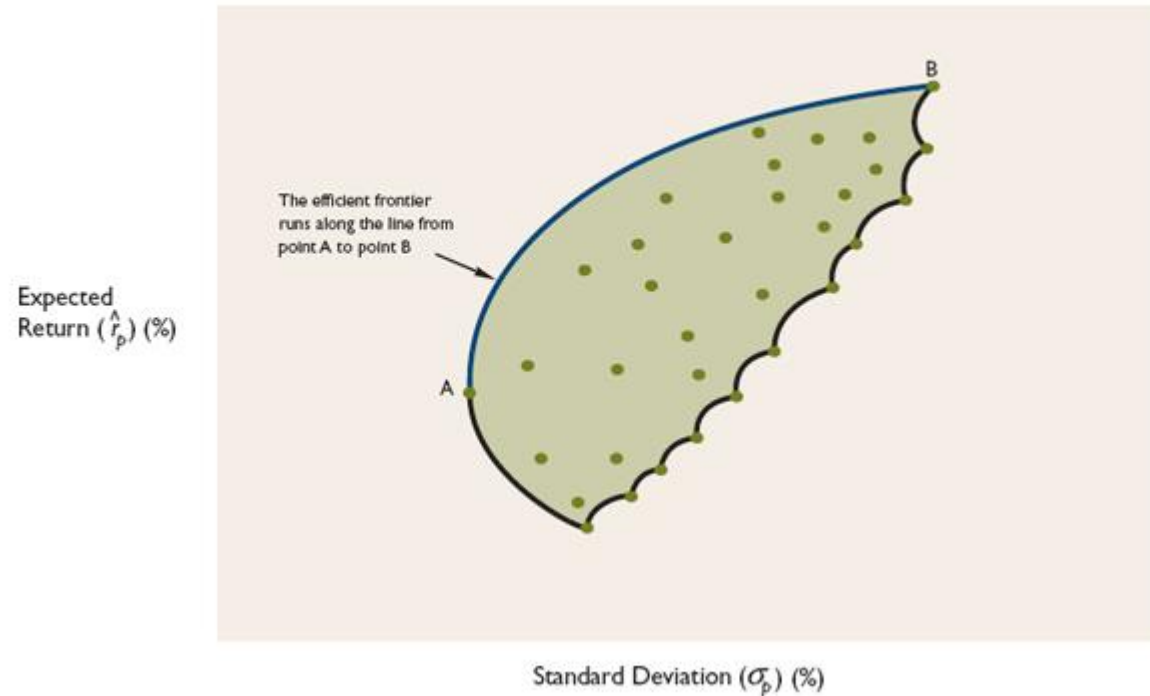


Asset B has a reward-to-risk ratio of **6.67 percent**, less than the **7.5 percent** offered by Asset A.

Portfolio Expected Returns and Betas for Both Assets

EFFICIENT PORTFOLIOS AND THE CAPITAL MARKET LINE

Portfolio Opportunity Set

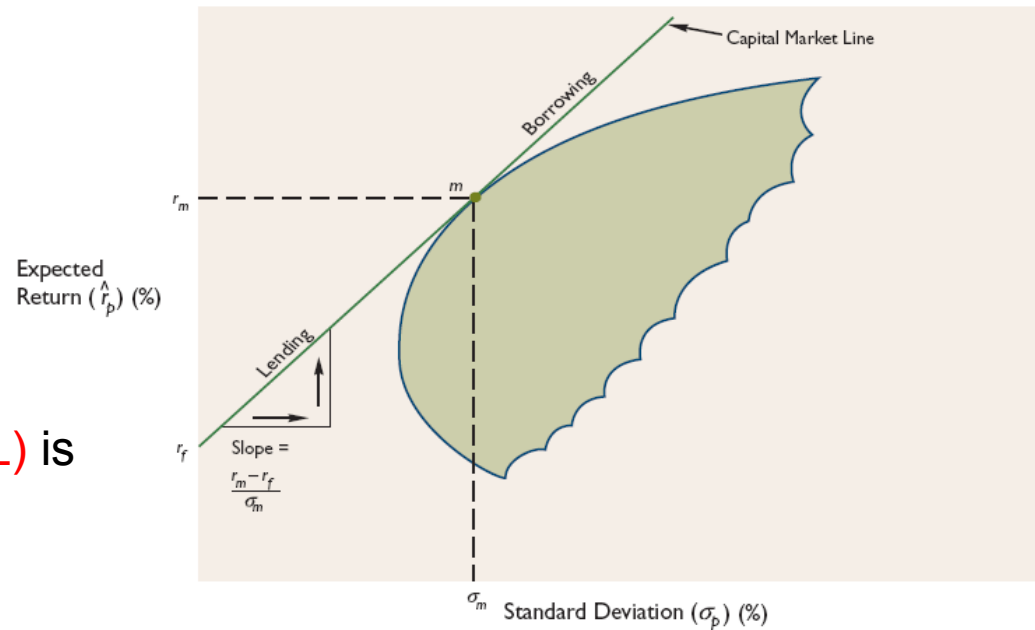


EFFICIENT FRONTIER

- If investors are able to borrow and lend money at the risk-free rate r_f , they can obtain any combination of risk and expected return on the straight line joining r_f and portfolio m

Equation of the **capital market line (CML)** is

$$\hat{r}_P = \hat{r}_f + \left(\frac{r_m - r_f}{\sigma_m} \right) \sigma_P$$



LARGE ASSET PORTFOLIO

- For example, whereas 45 correlation coefficients are needed for a portfolio containing 10 securities, 4,950 correlation coefficients must be computed for a portfolio containing 100 securities. In other words, a 10-fold increase in securities causes a greater than 100-fold increase in the required calculations.
- In addition, a **substantial computational undertaking** is required to find the particular portfolio of securities that minimizes portfolio risk for a given level of return or maximizes return for a given level of risk, even for a portfolio that contains only a few securities.

SECURITY MARKET LINE

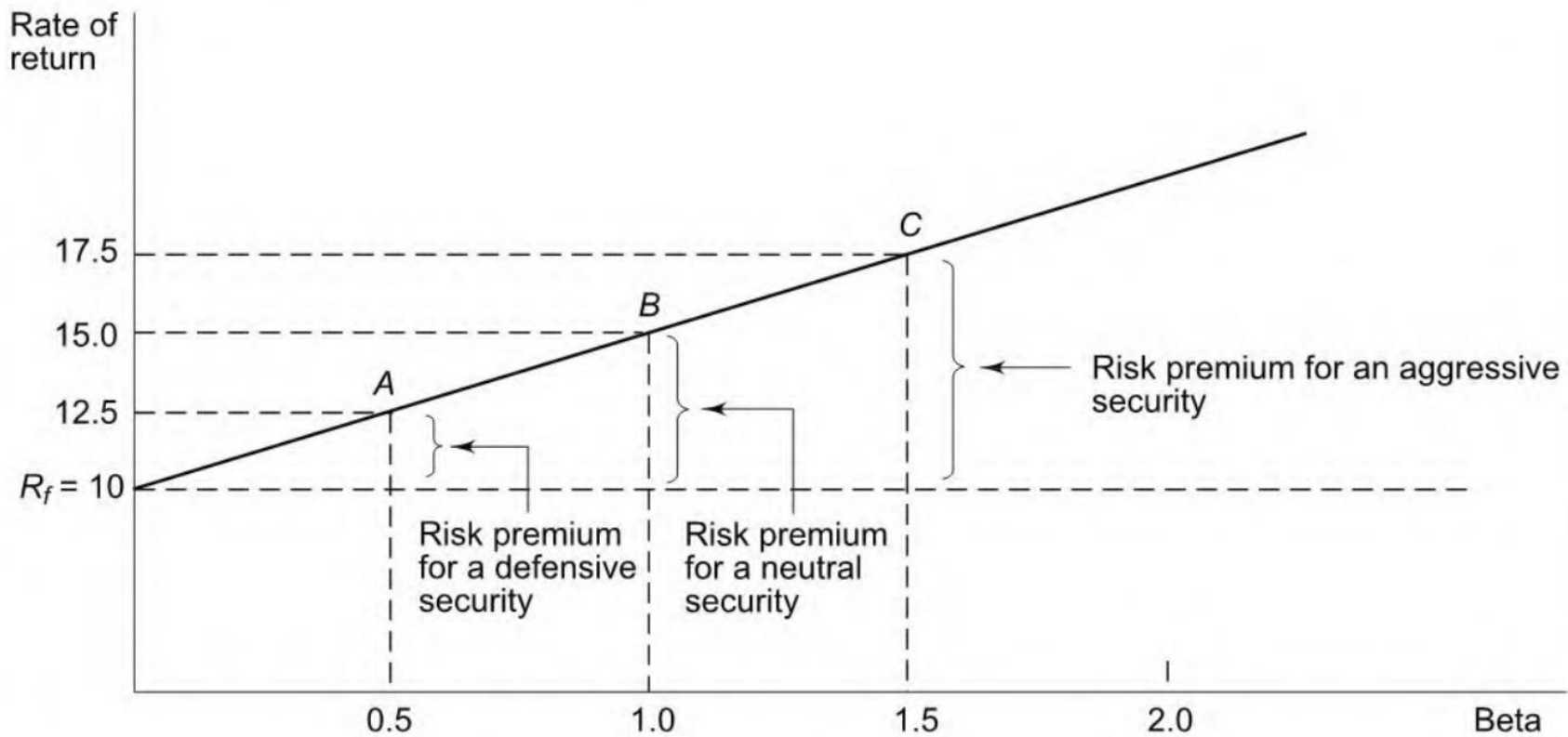
- This line, which we use to describe the relationship between **systematic risk and expected return** in financial markets, is usually called the security market line (SML).
- Because **all the assets in the market must plot on the SML, so must a market portfolio made up of those assets.**

the slope of the SML as:

$$\text{SML slope} = \frac{E(R_M) - R_f}{\beta_M} = \frac{E(R_M) - R_f}{1} = E(R_M) - R_f$$

The term $E(R_M) - R_f$ is often called the **market risk premium**

SML INTERPRETATION



IDEAL WORLD: RISK AND REWARD

Assets A (7.5%) and B (6.67%) could not persist in a well-organized, active market, because **investors would be attracted to Asset A and away from Asset B**. As a result, Asset A's price would rise and Asset B's price would fall. Because prices and returns move in opposite directions, A's expected return would decline and B's would rise.

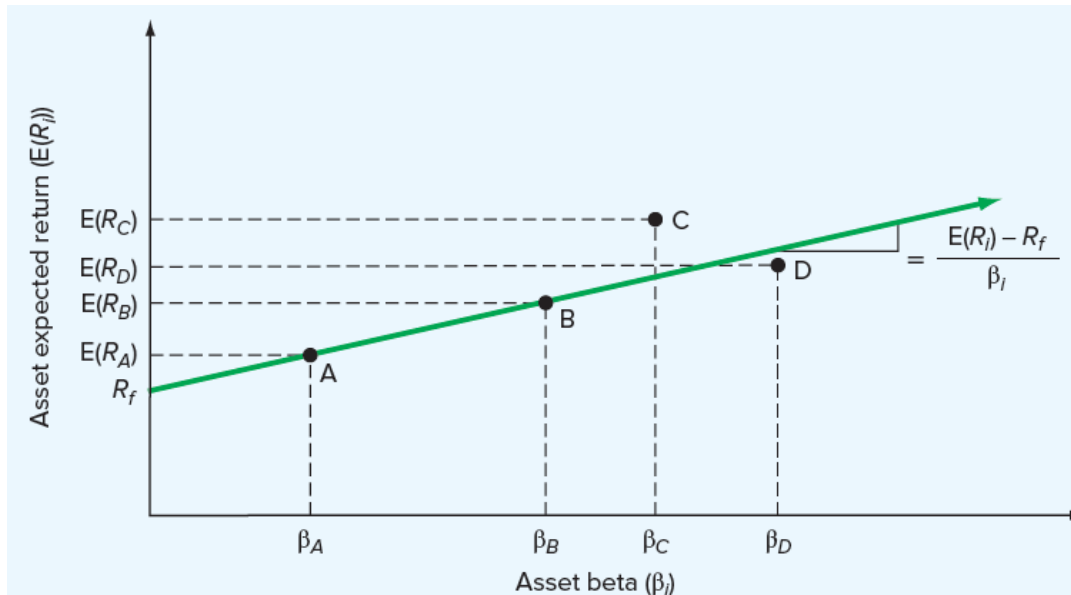
The reward-to-risk ratio must be the same for all the assets in the market.

This buying and selling would continue until the two assets (A and B) plotted on exactly the same line, which means they would offer the same reward for bearing risk. In other words, in an active, competitive market, we must have the situation that:

$$\frac{E(R_A) - R_f}{\beta_A} = \frac{E(R_B) - R_f}{\beta_B}$$

This is the fundamental relationship between risk and return.

EXPECTED RETURN AND RISK



The fundamental relationship between beta and expected return is that all assets must have the same reward-to-risk ratio, $[E(R_i) - R_f] / \beta_i$. This means that they would all plot on the same straight line. Assets A and B are examples of this behavior. Asset C's expected return is too high; Asset D's is too low.

Expected Returns and
Systematic Risk

Buy Low, Sell High

An asset is said to be overvalued if its **price is too high given its expected return and risk.**

Suppose you observe the following situation:

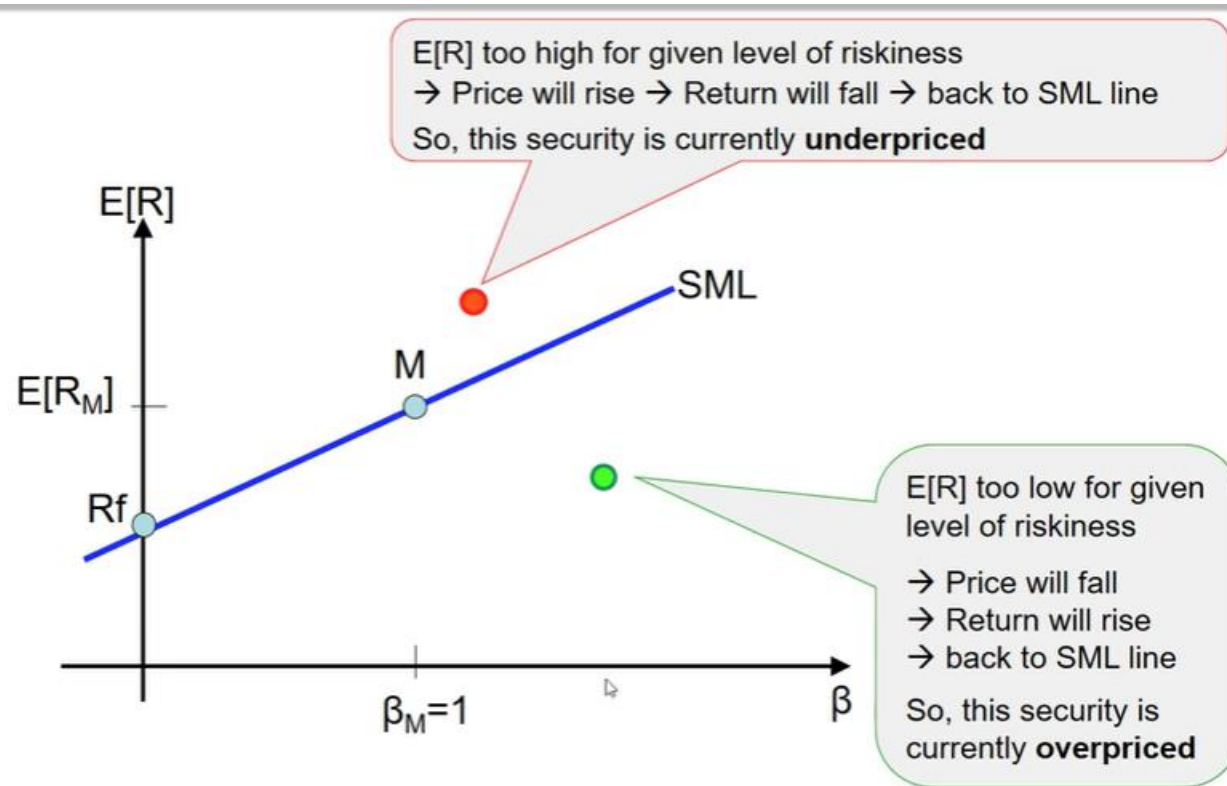
Security	Beta	Expected Return
SWMS Co.	1.3	14%
Insec Co.	.8	10

The risk-free rate is currently 6 percent. **Is one of the two securities overvalued relative to the other?**

For SWMS, this ratio is $(14\% - 6\%) / 1.3 = 6.15\%$. For Insec, this ratio is 5 percent.

Insec Co. is overvalued relative to SWMS Co.

UNDER/OVERPRICED STOCK



CAPITAL ASSET PRICING MODEL (CAPM)

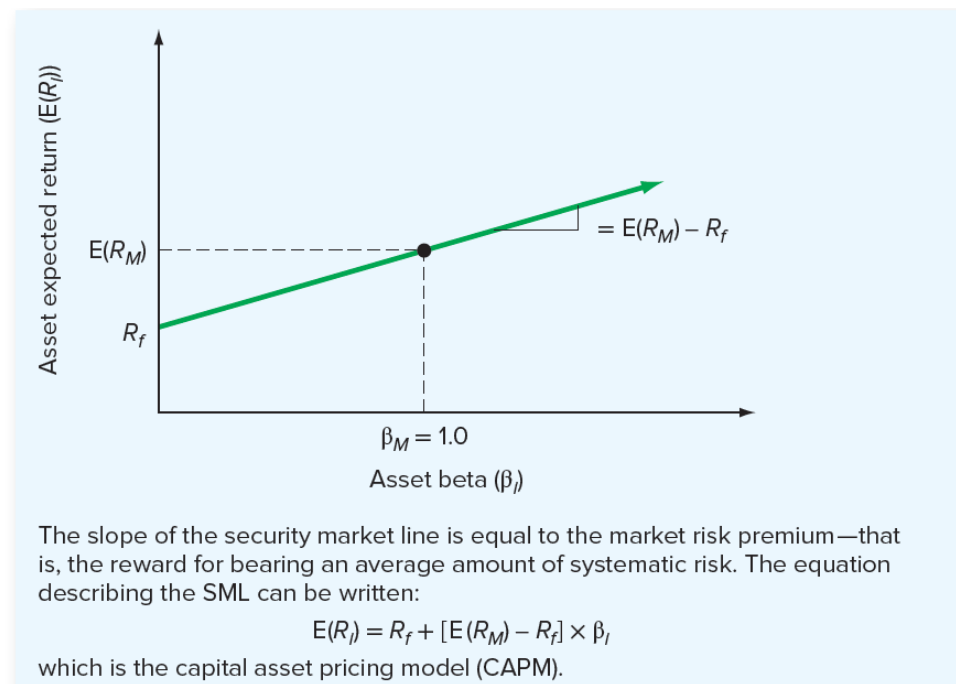
- We know that its reward-to-risk ratio is the same as the overall market's:

$$\frac{E(R_i) - R_f}{\beta_i} = E(R_M) - R_f$$

If we rearrange this, then we can write the equation for the SML as:

$$E(R_i) = R_f + [E(R_M) - R_f] \times \beta_i$$

This result is the famous **capital asset pricing model (CAPM)**.



ABOUT CAPM

The CAPM shows that the expected return for a particular asset depends on three things:

1. *The pure time value of money*: As measured by the risk-free rate, R_f , this is the reward for merely waiting for your money, without taking any risk.
2. *The reward for bearing systematic risk*: As measured by the market risk premium, $E(R_M) - R_f$, this component is the reward the market offers for bearing an average amount of systematic risk in addition to waiting.
3. *The amount of systematic risk*: As measured by β_i , this is the amount of systematic risk present in a particular asset or portfolio, relative to that in an average asset.

ASSUMPTIONS AND LIMITATIONS OF THE CAPM

- Investors hold **well-diversified portfolios** of securities. Hence, their return requirements are influenced primarily by the **systematic** (rather than total) risk of each security.
- Securities are **traded actively** in a competitive market, where information about a given firm and its future prospects is freely available.
- Investors can **borrow and lend at the risk-free rate**, which remains constant over time.
- There are **no brokerage charges** for buying and selling securities.
- There are **no taxes**.
- All investors prefer the security that provides the **highest return for a given level of risk** or the lowest amount of risk for a given level of return.
- All investors have common (**homogeneous**) **expectations** regarding the expected returns, variances, and correlations of returns among all securities.

RELATIONSHIP BETWEEN RISK AND RETURN

- What is the relationship between the risk of a security, as measured by its beta, and its expected return?
- **Security Market Line**
 - According to the CAPM, risk and return are related in a linearly:

$$E(R_j) = R_f + \beta_j [E(R_M) - R_f]$$

$E(R_j)$ is the expected return on security j

R_f is the risk-free return

β_j is the beta

$E(R_M)$ is the expected return on the market portfolio

COMPREHENSIVE PROBLEM

- The risk free rate is 4%, and the required return on the market is 12%.
 - What is the required return on an asset with a beta of 1.5?
 - What is the reward/risk ratio?
 - What is the required return on a portfolio consisting of 40% of the asset above and the rest in an asset with an average amount of systematic risk?

SOLUTION

- $R = .04 + 1.5 \times (.12 - .04) = .16$ (Use CAPM)
- The reward/risk ratio is 8% $[(.16 - .04) / 1.5]$
- $R = (.4 \times .16) + (.6 \times .12) = .136$

PROBLEM

- Suppose the risk-free rate is 4 percent, the market risk premium is 8.6 percent, and a particular stock has a beta of 1.3. Based on the CAPM, what is the expected return on this stock? What would the expected return be if the beta were to double?

SOLUTION

- With a beta of 1.3, the risk premium for the stock is $1.3 \times 8.6\%$, or 11.18 percent. The riskfree rate is 4 percent, so the expected return is 15.18 percent.
- If the beta were to double to 2.6, the risk premium would double to 22.36 percent, so the expected return would be 26.36 percent

PROBLEM FOR STUDENTS

Expected Return and Standard Deviation This problem will give you some practice calculating measures of prospective portfolio performance. There are two assets and three states of the economy:

State of Economy	Probability of State of Economy	Rate of Return If State Occurs	
		Stock A	Stock B
Recession	.20	−.15	.20
Normal	.50	.20	.30
Boom	.30	.60	.40

What are the expected returns and standard deviations for these two stocks?

SOLUTION

The expected returns are just the possible returns multiplied by the associated probabilities:

$$E(R_A) = (.20 \times -.15) + (.50 \times .20) + (.30 \times .60) = .25, \text{ or } 25\%$$

$$E(R_B) = (.20 \times .20) + (.50 \times .30) + (.30 \times .40) = .31, \text{ or } 31\%$$

The variances are given by the sums of the squared deviations from the expected returns multiplied by their probabilities:

$$\begin{aligned}\sigma_A^2 &= .20 \times (-.15 - .25)^2 + .50 \times (.20 - .25)^2 + .30 \times (.60 - .25)^2 \\ &= (.20 \times -.40^2) + (.50 \times -.05^2) + (.30 \times .35^2) \\ &= (.20 \times .16) + (.50 \times .0025) + (.30 \times .1225) \\ &= .0700\end{aligned}$$

$$\begin{aligned}\sigma_B^2 &= .20 \times (.20 - .31)^2 + .50 \times (.30 - .31)^2 + .30 \times (.40 - .31)^2 \\ &= (.20 \times -.11^2) + (.50 \times -.01^2) + (.30 \times .09^2) \\ &= (.20 \times .0121) + (.50 \times .0001) + (.30 \times .0081) \\ &= .0049\end{aligned}$$

The standard deviations are thus:

$$\sigma_A = \sqrt{.0700} = .2646, \text{ or } 26.46\%$$

$$\sigma_B = \sqrt{.0049} = .07, \text{ or } 7\%$$

PROBLEM FOR STUDENTS

- Suppose the risk-free rate is 8 percent. The expected return on the market is 16 percent. If a particular stock has a beta of .7, what is its expected return based on the CAPM? If another stock has an expected return of 24 percent, what must its beta be?

SOLUTION

Because the expected return on the market is 16 percent, the market risk premium is $.16 - .08 = .08$, or 8%. The first stock has a beta of .7, so its expected return is $.08 + .7 \times .08 = .136$, or 13.6%.

For the second stock, notice that the risk premium is $.24\% - .08 = .16$, or 16%. Because this is twice as large as the market risk premium, the beta must be exactly equal to 2. We can verify this using the CAPM:

$$\begin{aligned} E(R_i) &= R_f + [E(R_M) - R_f] \times \beta_i \\ .24 &= .08 + (.16 - .08) \times \beta_i \\ \beta_i &= .16 / .08 \\ &= 2.0 \end{aligned}$$

CAPM IN RESEARCH

To derive the market model, we start with the CAPM:

$$E(R_i) = R_f + \beta[E(R_M) - R_f]$$

Since CAPM is an equation, we can subtract the risk-free rate from both sides, which gives us:

$$E(R_i) - R_f = \beta[E(R_M) - R_f]$$

This equation is deterministic, by which we mean it is exact. In a regression, we realize that there is some indeterminate error. We need to formally recognize this in the equation by adding epsilon, ϵ , which represents this error:

$$E(R_i) - R_f = \beta[E(R_M) - R_f] + \epsilon$$

Finally, think of the above equation in a regression. Since there is no intercept in the equation, the intercept is zero. However, when we estimate the regression equation, we can add an intercept term, which we will call alpha:

$$E(R_i) - R_f = \alpha_i + \beta[E(R_M) - R_f] + \epsilon$$

This equation is often called the “market” model, though it is not the only equation with that name, which is a source of confusion. The intercept term is known as Jensen’s alpha, and it represents the “excess” return. If CAPM holds exactly, this intercept should be zero.

EXTENSION - THREE AND FIVE FACTORS REGRESSION MODELS

$$R_{it} - R_{Ft} = a_i + b_i(R_{Mt} - R_{Ft}) + s_iSMB_t + h_iHML_t + e_{it}.$$

Eugene Fama F., French, Kenneth R., (1993), "Common Risk Factors in the Return on Stocks and Bonds," *Journal of Financial Economics*, 33, pp. 3–56.

$$R_{it} - R_{Ft} = a_i + b_i(R_{Mt} - R_{Ft}) + s_iSMB_t + h_iHML_t + r_iRMW_t + c_iCMA_t + e_{it}.$$

Fama, E. F., & French, K. R. (2015). A five-factor asset pricing model. *Journal of financial economics*, 116(1), 1-22.

In this equation R_{it} is the return on security or portfolio i for period t , R_{Ft} is the riskfree return, R_{Mt} is the return on the value-weight (VW) market portfolio, SMB_t is the return on a diversified portfolio of small stocks minus the return on a diversified portfolio of big stocks, HML_t is the difference between the returns on diversified portfolios of high and low B/M stocks, and e_{it} is a zero-mean residual.

In this equation RMW_t is the difference between the returns on diversified portfolios of stocks with robust and weak profitability, and CMA_t is the difference between the returns on diversified portfolios of the stocks of low and high investment firms, which we call conservative and aggressive.

END OF TOPIC